



Omar Marghadi

Date of birth: 15/03/2003 | Nationality: Moroccan | Phone: (+33) 0751014124 (Mobile) |
Email address: omar3.marghadi@gmail.com | LinkedIn:
<https://www.linkedin.com/in/omar-marghadi/> | Address: Paris, France (Home) | "Humans
create Meaning; Machines create Possibilities."

About me

Industrial engineering graduate passionate about AI, robotics, and autonomous systems. Focused on perception, sensor fusion, and intelligent control.

Education & Training

Master in Engineering (M2) : track Data and Artificial Intelligence in Industrial Systems | École Centrale de Lyon | 02/09/2024 - 03/10/2025 | Lyon, France

Final grade: Magna Cum Laude | **Level in EQF** 7 | **Thesis:** Multi-Floor Spatial Perception and Anomaly Detection for Domestic Service Robots and Contextualized Alert Generation | **Link:** <https://offre-formation.ec-lyon.fr/en/node/45695277>

Industrial Engineer: AI & Data Science Track | Arts & Metiers (ENSAM) | 01/09/2020 - 03/10/2025 | Meknes, Morocco

Main subjects: Industrial Computing & Programming - Automatic Control & Mechatronics - Applied Mathematics & Numerical Methods - Mechanical & Computer-Aided Design (CAD/CAM/CAE) - Materials Science & Manufacturing Processes

Level in EQF 7 | **Thesis:** Completed at Centrale Lyon as part of double degree program

Baccalaureate in Mathematics and Engineering Science - Highest Honors

Recommendations

Associate Professor, Mathematics and Computer Science Department, ENSAM-Meknes | Mohamed HOSNI, Eng, PhD | (+212) 0665702770 | m.hosni@umi.ac.ma

Associate Professor, Director of Master's Program- École Centrale de Lyon - LTDS UMR 5513 | Lyes Nechak, HDR | (+33) 0472186481 | lyes.nechak@ec-lyon.fr

Work experience

AI Robotics Research engineer | Orange Innovation | 01/04/2025 - 30/09/2025 | Lannion, France

- Novel multi-floor map alignment algorithm achieving 73.8% IoU using morphological preprocessing
- Sensor fusion framework integrating WiFi quality metrics and environmental IoT sensors with Roborock vacuum robots LIDAR SLAM navigation in unstructured home environments
- Graph Attention Network (GAT) architecture for spatiotemporal anomaly detection (+11% F1-score vs. statistical baselines) for ROS2
- Edge-optimized Small Language Model for real-time contextualized alert generation (<200ms inference)

Research Intern GenAI & Human-AI Collaboration | Capgemini Engineering | 25/06/2024 - 25/08/2024 | Casablanca, Morocco

- Developed multi-agent AI system to assist Capgemini engineers in daily requirement analysis tasks
- Architected LangGraph-based framework enabling human-AI collaboration through specialized AI tools
- Engineered ETL pipeline with domain-specific parsers for automated engineering document processing

Computer Vision Research Intern | OCP Group | 01/07/2023 - 01/08/2023 | Morocco

- Vision-based operator pose estimation using HRNet for real-time ergonomics monitoring in industrial environments

Projects

Drone Defense Hackathon : Delivery Drone Perception in Crisis Zones | 15/09/2025 - Current

- Developing real-time perception and sensor fusion for autonomous delivery drones in incertain environments
- AirSim-based simulation for obstacle & Threat Detection and navigation scenarios (camera, radar)

<https://www.agorize.com/en/challenges/drone-defense-hackathon/>

Bio-mimic Seal Robot: Ending Seal Over-Tagging for Ocean Current Mapping and Climate Monitoring

- Designed a bio-inspired mechanical structure replicating seal locomotion for efficient underwater propulsion
- Implemented a three-tier embedded control architecture (WiFi LoRa32 + OLED ESP8266 relay + Arduino servo control via ESP-NOW)
- Integrated multi-sensor: temperature (DS18B20), tilt (SCA100T-D01), IMU/gyro (MPU-6050), enabling sensor fusion for accurate orientation and motion tracking

<https://drive.google.com/drive/folders/1-7iPqGs-j7LXw5WinUddJzxKhef7BzuB>

AI-Based Robotic Control Systems: Literature Review and Paper Writing

Authored a 25-page literature review and research paper on Deep RL & Imitation Learning for robotics, featuring case studies, formal models, and quantitative performance benchmarks.

<https://drive.google.com/drive/folders/1oDj0JzerlpoA025OXubkYxglabPYDnJk>

Industrial Vision System for counting on production line - Industry Collaboration

Addressed complex challenges: overlapping/entangled bars, variable positioning, and dynamic conditions of steel bars on production lines using Adaptive Vision software.

Language Skills

Mother tongue(s): Arabic

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C2	C2	C1	C1	C1
French	C2	C2	C1	C1	C1

Skills

Technical | Programming: C, C++, python, javascript, embedded system | Experience with Robot Operating System (ROS) and Linux (ROS1 and ROS2) | Good knowledge of Pytorch and related Python libraries (scikit-learn, Numpy, Pandas etc.). | knowledge of Matlab Simulink, ROS/Gazebo 7 framework, Yolo and OpenCV | Unity 3D Knowledge | Sensor Fusion | Computer VIsion | Control system | Robot Perception | Neural Networks | Intelligent Systems

Research | Research Methodology | SLR (Systematic Literature Review) | Technical & Non-Technical Writing | Bibliometric Techniques

Soft | Time Management, Attentiveness, Problem solving, Effective communication skills | Having interpersonal communication skills | Teamwork skills(conflict management and resolution team building and collaboration)

Declaration